

End-to-End Data Analytics Project: Supply Chain Performance Analysis

Project Overview

This project analyzes supply chain and shipping operations to identify factors affecting operational efficiency, customer satisfaction, and business performance. Using Python, SQL, Excel, and Tableau, the project follows a complete analytics workflow from data preparation and exploration to dashboard development and business recommendations.

Business Problem

Which product categories generate the highest revenue? How do inventory levels affect performance? Which suppliers have the highest defect rates? Do higher transportation costs improve outcomes? How does delivery performance impact customer satisfaction?

Dataset Information

Supply Chain Dataset: product categories, revenue, inventory, shipping times, transportation costs, supplier performance, and defect rates. E-Commerce Shipping Dataset: shipment methods, customer ratings, delivery performance, warehouse operations, and customer service interactions.

Tools & Technologies

Excel, Python (Pandas), SQL, and Tableau.

Data Preparation

Removed duplicates, handled missing values, standardized columns, validated numerical fields, corrected category labels, reviewed outliers, and created performance metrics.

Exploratory Data Analysis

Revenue analysis, inventory analysis, transportation analysis, supplier performance analysis, and customer satisfaction analysis were performed to identify business insights.

SQL Analysis

Revenue by product category, average shipping time by transportation mode, supplier defect comparisons, inventory availability analysis, and KPI reporting.

Dashboard Development

Interactive Tableau dashboard including Revenue by Location, Revenue by Product Category, Monthly Revenue Trends, Revenue vs Defect Rate, and Defect Rate by Location.

Key Findings

Revenue concentration in a few categories, inventory imbalances, supplier quality impacts performance, transportation costs require optimization, and delivery performance drives customer satisfaction.

Business Recommendations

Improve inventory forecasting, monitor supplier performance, optimize logistics costs, improve delivery performance, and track KPIs through dashboards.

Conclusion

This project demonstrates an end-to-end analytics workflow using Python, SQL, Excel, and Tableau to generate actionable business insights and support data-driven decision making.